Integrating Ventilation Air with a VRF System

Chris Soh
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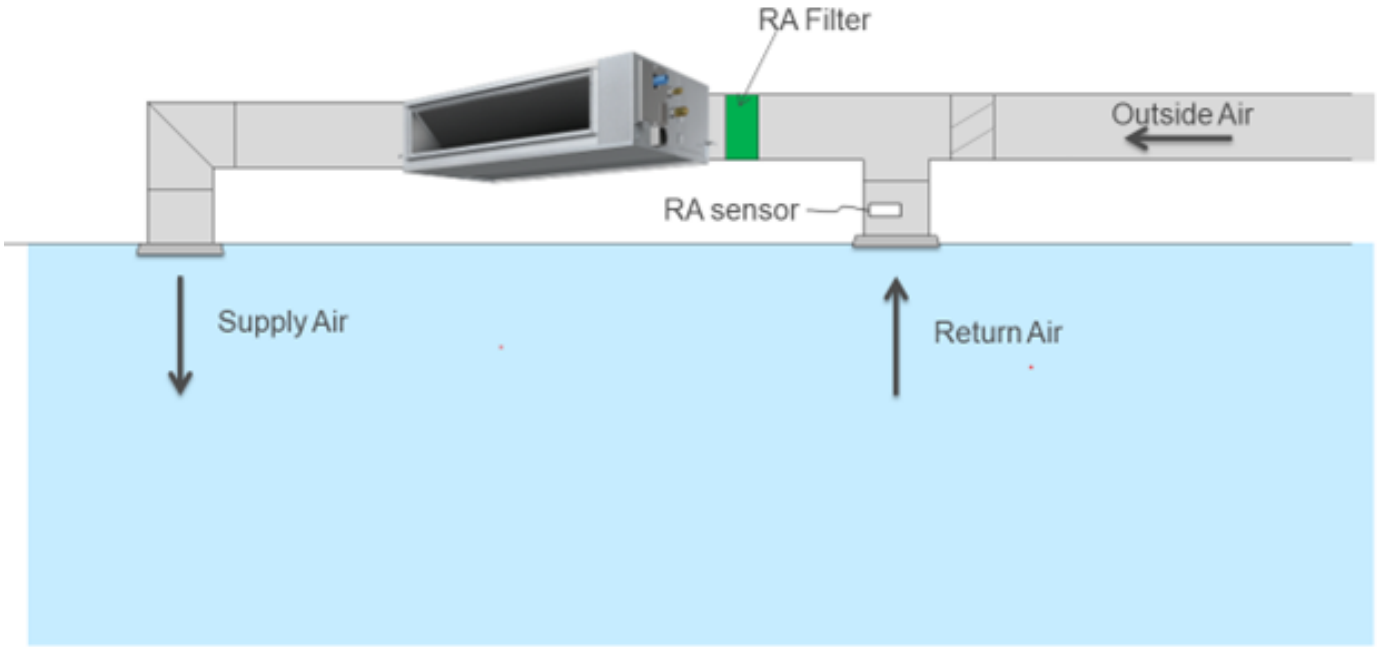
*The ventilation options for the V*ariable Refrigerant Flow (VRF) *system* for compliance with ASHRAE 62.1 has always been a subject of great interest, now more so than ever. Not only is it of great importance in designing systems for new buildings, but also in evaluating system effectiveness in an effort to reopen businesses and schools following the COVID pandemic. Building owners and managers are being asked to review outdoor airflow rates and compare them to the most current version of ASHRAE 62.1 in their state. This article discusses how to integrate the ventilation air required of 62.1 with a VRF system.

Ventilation Air Strategy

The 2020 ASHRAE handbook- HVAC System and Equipment ch18 “Variable Refrigerant Flow” identifies three common ventilation air strategies used by VRF Systems — direct, integrated, and decoupled. Both the direct and integrated strategies use the heating/cooling duct of the VRF system to supply fresh air, whereas in the decoupled strategy, the ventilation duct is dedicated to deliver fresh air to all occupied zones and is separate from the heating/cooling duct. Each of these strategies has its own set of pros and cons.

The Direct Method

In the direct method, outdoor air is mixed with return air and conditioned by the VRF indoor unit. This outdoor air is ducted into the return air duct of the VRF indoor unit. If a ductless indoor unit is used, be aware that the manufacturer often restricts the OA amount to between 3-10% of the unit airflow rate, due to the size of the fresh air connection and the concern of noise.



Application Considerations:

- Consider the entering air temperature at the VRF unit’s coil. Typically, the allowable operating ranges for standard VRF fan coils are 57° WB to 77° WB in cooling operation and 59° DB to 80° DB in heating operation. Designers must ensure the air temp entering the VRF unit’s coil does not exceed these limits for prolonged periods to avoid equipment shutdown or errors.

- The mixed-air temperature will impact unit performance. The cooling coil rating of VRF indoor units at AHRI conditions is based on 80° DB/67° WB and 70° DB for heating. If the air entering the coil deviates from these temperatures, the coil's performance will deviate from the catalog ratings. For cooling, lowering the entering air temperature reduces the coil capacity and in heating will raise the coil capacity. This is a function of the log mean temperature difference between the refrigerant and air temperatures across the coil.

- Verify that the VRF indoor unit supply (discharge air) temperatures are at the level required to maintain desired indoor conditions (temperature and humidity). ASHRAE handbook recommends the minimum supply air temperature from the ventilation system should be 59-63F.

- To ensure minimum ventilation requirements, the VRF indoor unit fan must run at design fan speed at all times.

The direct method is typically the cheapest design option as there is no additional equipment and separate ventilation air duct systems to pretreat and distribute the outdoor air. The downside to the direct method is that untreated outside air will be delivered to the space with no heating or cooling until the space thermostat calls for it. This is due to the heating and cooling coils being activated by space temperature. If too much humid outdoor air is pulled into the space it can cause occupants to feel clammy and uncomfortable. A direct method is best used for application in mild to moderate climates, and or/situations requiring small amounts of outdoor air.

The Integrated Method

In the integrated method, outdoor air is pretreated by mechanical means such as a 100% outdoor air unit, a dedicated outside air system (DOAS) or energy recovery ventilator (ERV), before being supplied to the VRF indoor unit through the return air or supply ductwork. The integrated method ensures that the air being introduced into the space is not two different temperatures, as to not increase the complexity of the thermal management within the zone.

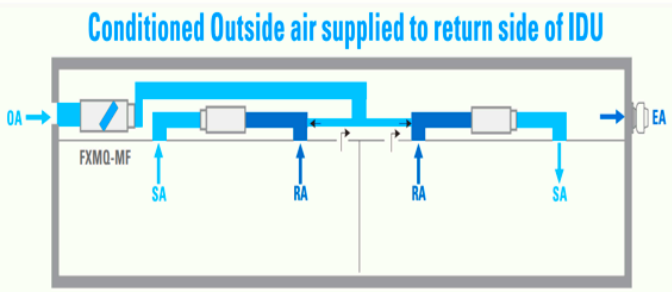
Application considerations:

- VRF indoor units provide final tempering of the mixed airstream and should be sized accordingly.

- In part load operation, the VRF indoor unit's supply air temperature is reduced to satisfy a reduction in the room sensible cooling load. This has a direct impact on unit latent

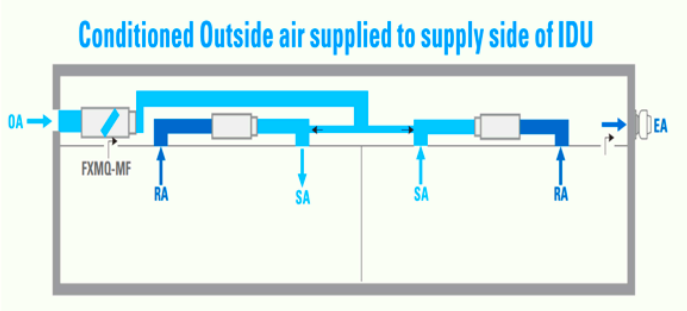
performance and should be considered, along with climate conditions, when determining how much pretreatment is required.

- To ensure minimum ventilation requirements, the VRF indoor unit fan must run at design fan speed at all times.



Notes

- If the conditioned outdoor air is delivered at a cold temperature rather than a neutral temperature, it results in cool air entering the coil in the IDU. This must be considered during equipment selection, as it may impact IDU capacity



Notes

- If the outdoor air is delivered at a cold temperature, rather than reheated to near space temperature, this configuration can offer the opportunity to downsize the IDU. However, measurement and balancing are usually more difficult in this scenario than when the outdoor air is delivered directly to the space.

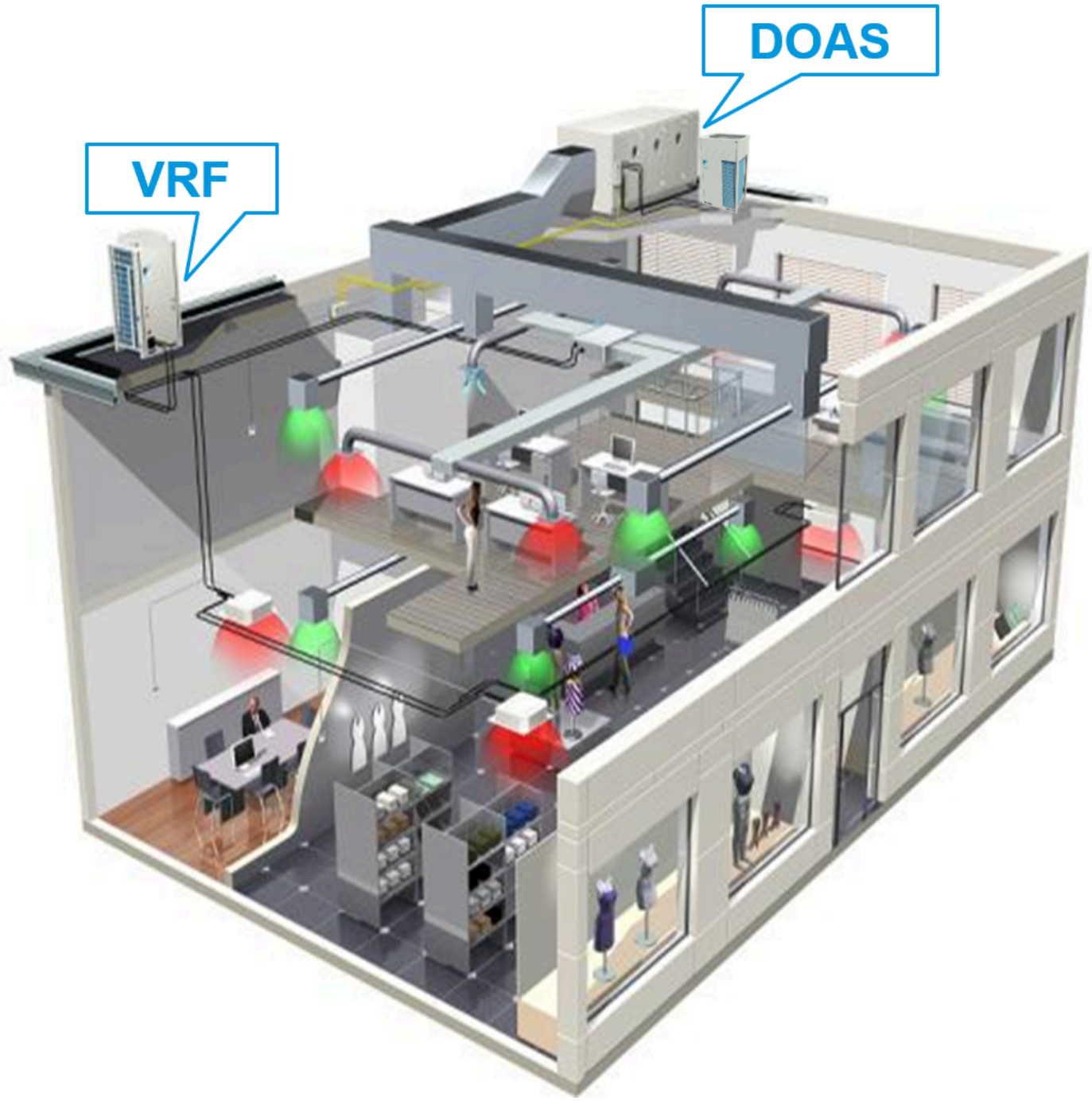
The primary advantage of the integrated method is that it reduces cost and space needed to install additional ductwork and separate diffusers. The ventilation equipment lowers the OA temperature entering the VRF indoor unit which increases the ratio of fresh air allowable, that in turn increases the application flexibility of the VRF system to satisfy the ventilation requirement of large spaces with high occupancy. Compared to the Direct Method, the Integrated method yields a smaller capacity indoor unit selection and better comfort/performance especially at part load operation.

To help improve humidity performance of the VRF system,

- Schedule dry mode during afterhours. This will help dehumidify the building while its unoccupied.

- Reduce the introduction of OA when not needed, this can be done by implementing CO2 sensors. This is especially recommended for areas with varying occupancy levels such as with a conference room.

The Decoupled Method



In the decoupled method, ventilation air is supplied directly to the conditioned space. Both the dedicated ventilation system, such as a DOAS and a VRF system can be installed. The DOAS conditions the outdoor air based on dew point control, to deliver neutral air and the function of the VRF system is to maintain comfort in the conditioned space. Since no mixing is involved, ventilation rate delivery to the space/zone can easily be verified and continuously monitored.

Application Considerations:

- The DOAS must be designed properly so that incoming outdoor air does not cause short circuiting issues with return air of the VRF indoor units, or influence temperature sensors.

- Fans constitute a significant portion of the energy consumption for DOAS. Decentralizing a DOAS into multiple, smaller split systems helps to lower the design external static pressure requirement and fan energy use. Such systems can also provide additional benefits such as:

* Zone Ventilation Control

- Control Different occupancy schedules (More than one unit can be off)

- Condition the OA to different indoor conditions

* Remote condensing unit provides installation flexibility

- Structural weight and size implications

- Serviceability/maintenance – condensing units can be in one location

* Smaller ducts and ceiling plenums save on installation costs. Smaller ducting is also easier to manage in retrofit and existing HVAC installations.

Benefits of the decoupled method include the ability to introduce any amount of OA

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The VRF indoor unit fan can be programmed to cycle on/off based on heating and cooling requirements, regardless of ventilation demands, thus increasing system efficiency. The main drawback of this configuration is that it requires the installation of additional ductwork and separate diffusers, resulting in added material first cost. It can be challenging to find an optimum location for this diffuser while still maintaining good air distribution for the local cooling/heating diffuser.

Conclusion

Designers can select many different types of ventilation equipment to compliment a VRF system using the strategies discussed in this article. The article provides you with all the information needed to choose the best strategy for your project taking into

consideration the climate, and the application. It is important to note that no single approach is best for every application. When it comes to building readiness for reopening following COVID, it is best to evaluate and re-commission your VRF systems as needed to ensure building is operating properly per original design intent.

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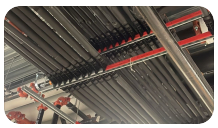

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